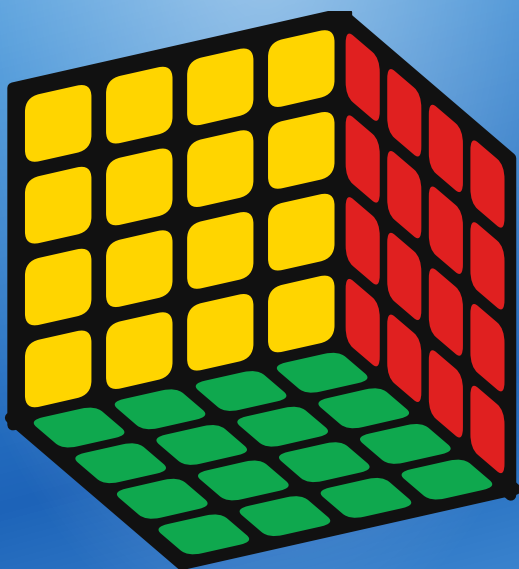


4×4 CUBE

6 SIDES · 24 CENTERS · 24 EDGES · 8 CORNERS

SOLUTION GUIDE



SOLVE IT LIKE A 3×3!

STAGE 1:

GET TO KNOW YOUR 4×4 CUBE

THE BIG IDEA

A 4×4 has no fixed centers and its edges are split into **two halves**. The trick is **reduction**: first build the center blocks and join the edge halves, and then the whole puzzle behaves **exactly like a 3×3** — so you finish it with the same moves you already know.

THE PARTS:

CORNER PIECES

Three (3) colors. There are **eight (8)**, one at each corner of the cube — just like a 3×3.



EDGE PIECES (WINGS)

Two (2) colors. There are **24**, in matching pairs. Two wings of the same two colors make one 3×3 edge once you pair them.



CENTER PIECES

One (1) color, four to a face (**24** total). They are **not** fixed — you build each face's 2×2 center block yourself.



WHITE and **YELLOW** are on opposite faces. We build the white side first and finish on the yellow side. (Use YOUR cube's colors; the steps are the same.)

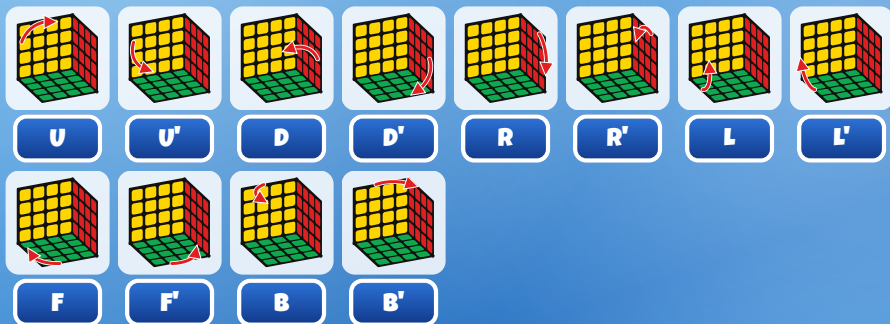
EACH FACE TURN HAS A LETTER

A letter alone (like **(R)**) means turn that face a quarter turn **clockwise** as you look straight at it. A letter with an apostrophe (**(R')**) means **counter-clockwise**. A letter with a **2** (**R2**) means a half turn. **U**p, **D**own, **R**ight, **L**eft, **F**ront, **B**ack.

For the very last "parity" page you will also use **wide** turns, written like **Rw** — turn the face **and** the slice next to it together (two layers at once).

VERY IMPORTANT

Hold your cube to match each picture before you start a sequence. Dark gray in a picture means **that color does not matter**.



STAGE 2:

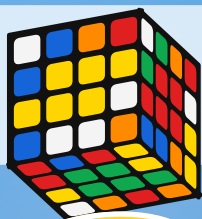
BUILD THE 6 CENTERS

HOLDING YOUR 4*4: Pick a color to start (we use white). Build its four center pieces into one 2x2 block on a face. Then do the same for every color — opposite colors go on opposite faces.

THE BAR METHOD: find two center pieces of your color that sit next to each other on a side face — that is a 2x1 **bar**. Turn an inner slice (the middle layer) to slide the bar up onto the face you are building. Make a second bar the same way and slide it up next to the first to finish the 2x2 block.



2R



**GOAL: SIX
SOLID CENTERS**

Here the top face already has one white bar; a second white bar waits in the front face's right column. **(2R)** (the inner slice next to the right face) slides it up to complete the white center.

TIPS:

- Build one face, then hold it on top or bottom so the inner slices you use for the next face do not break it.
- The last two colors are the tricky ones — that needs its own page (next).
- When all six centers are solid blocks, the centers are done — their colors now tell you where every other piece belongs.

STAGE 3:

THE LAST TWO CENTERS

HOLDING YOUR 4*4: When only two colors are left, they share the two opposite faces. Look at how their pieces are split, then use the matching fix below.

Hold the two unsolved faces on top and bottom. The four side centers are already solid — keep them that way. Only inner slices (and, for the diagonal case, a top/bottom turn) are used.

TWO COLUMNS SPLIT

If the swapped pairs line up as two columns, do:

2U2 2B2 2U2

(Row split? Use the mirror: 2U2 2L2 2U2)



2U2



2B2



2U2

DIAGONAL SPLIT

If the swapped pairs sit diagonal to each other, do:

U' 2R2 U' D' 2R2



U'



2R2



U'

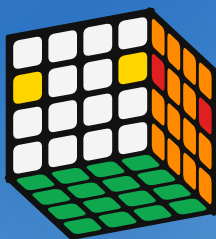
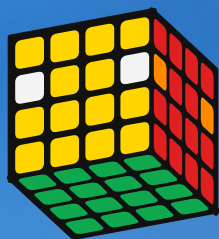


D'



2R2

Both unsolved faces finish at the same time:



TIPS:

- Turn only the inner slices for the column and row cases — the side centers stay solid.
- The diagonal fix uses one top and one bottom turn; that is fine, the side centers still come back solved.
- Done! All six centers are solid blocks — move on to pairing the edges.

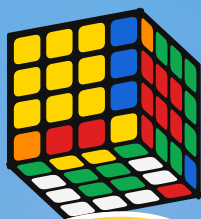
STAGE 4:

PAIR THE EDGES

HOLDING YOUR 4x4: Find two edge wings that share the same two colors and join them so they sit side by side as one edge. Repeat until all 12 edges are paired.

TIPS:

- Bring one wing to the top-front and its matching wing next to it, then turn a wide layer to lock them together — and turn it back so the centers stay solid.
- Keep finished pairs out of the way (in the top or bottom) while you join the next pair.
- Once all 12 edges are joined, STOP using slice and wide turns. From here, only turn whole faces — your 4x4 is now a 3x3!



**GOAL: 12
joined edges**

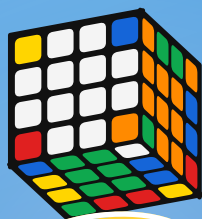
STAGE 5:

MAKE THE WHITE CROSS

HOLDING YOUR 4*4: Turn the cube so **WHITE** is on the bottom. Make a white plus on the bottom, and make each edge's side color match its center.

TIPS:

- Find a white edge, line it up under the matching side center, then turn that face twice to drop it into the cross.
- If a white edge is flipped, take it to the top first, then bring it down the right way.
- This is exactly the 3×3 white cross — only whole-face turns from now on.



GOAL: A WHITE CROSS

STAGE 6:

FINISH THE WHITE LAYER

HOLDING YOUR 4*4: Put the four white corners in place to complete the bottom layer, white on the bottom and side colors matching.

TIPS:

- Find a white corner in the top layer, hold it above the empty corner where it belongs.
- Use **(R) (U) (R)'** (and a top turn between tries) to drop it in with white on the bottom.



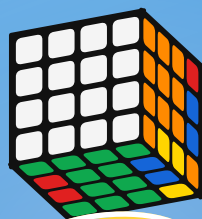
R



U



R'



**GOAL: FULL
WHITE LAYER**

STAGE 7:

SOLVE THE MIDDLE EDGES

HOLDING YOUR 4*4: Flip the cube so white is on the bottom and yellow on top. Place the four middle-layer edges (the ones with no yellow).

TIPS:

- Find a top edge with no yellow. Turn the top so its front color matches a side center, making a sideways T.
- If it belongs to the right, do **(U) (R) (U)' (R)' (U)' (F)' (U) (F)**. Mirror it for the left.



U



R



U'



R'



U'



F'



U



F

STAGE 8:

YELLOW CROSS & TOP

HOLDING YOUR 4*4: With yellow on top, make a yellow cross, then make the whole top face yellow.

TIPS:

- For the cross, repeat **(F) (R) (U) (R') (U') (F')** until the yellow plus appears.
- Then orient the yellow corners (turn the top so an unsolved corner is front-right and repeat a corner move) until the whole top is yellow.



F



R



U



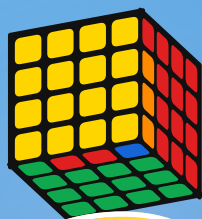
R'



U'



F'



**GOAL: YELLOW
TOP**

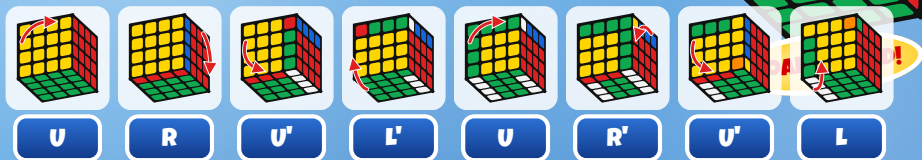
STAGE 9:

FINISH THE LAST LAYER

HOLDING YOUR 4×4: Move the last pieces into place: first the corners, then the edges, until the cube is solved.

TIPS:

- Position the yellow corners using a corner 3-cycle, turning the top between tries so the solved corners stay put.
- Then cycle the last edges into place. These are the same last-layer moves as the 3×3.



STAGE 10:

PARITY (4×4 ONLY)

Because a 4×4 has split edges, you can reach two cases a 3×3 never shows. They are not mistakes — just do the matching fix and keep going.

ONE FLIPPED EDGE (OLL)

If a single last-layer edge looks Flipped while making the yellow cross, do:

Rw2 B2 U2 Lw U2 Rw' U2 Rw U2 F2 Rw F2 Lw' B2 Rw2

TWO SWAPPED EDGES (PLL)

If only two last-layer edges need to swap at the very end, do:

2R2 U2 2R2 Uw2 2R2 Uw2



2R2



U2



U2



2R2



Uw2



2R2



Uw2

CONGRATULATIONS!

You reduced a 4×4 to a 3×3 and solved it! Centers, then edges, then the same moves you know. Now mix it up and do it again.

Every picture in this booklet was drawn by a computer model of the cube that checks every move really works.